5.9.1

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Construct a pair of tangents to a circle of radius 4 cm from a point P at a distance of 7 cm from its centre O, using matrix methods.
Solution:

Let the equation of the circle with centre $\mathbf{u}$ be

$$(\mathbf{x}-\mathbf{u})^\top(\mathbf{x}-\mathbf{u}) = r^2, \tag{1}$$

Given,

$$r = 4, \quad \|\mathbf{P}-\mathbf{u}\| = 7. \tag{2}$$

Let

$$\mathbf{u} = \mathbf{0}, \quad \mathbf{v} = \mathbf{P}-\mathbf{u}, \quad s = \mathbf{v}^\top\mathbf{v} = \|\mathbf{v}\|^2. \tag{3}$$

Define the orthonormal basis vectors as

$$\mathbf{u_1} = \frac{\mathbf{v}}{\sqrt{s}}, \qquad \mathbf{u_2} = \frac{\mathbf{J}\mathbf{v}}{\sqrt{s}} \tag{4}$$

$$\mathbf{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{5}$$

Let the unit normal to the tangent at the point of contact be expressed as

$$\frac{\mathbf{n}}{\|\mathbf{n}\|} = \alpha\mathbf{u_1} + \beta\mathbf{u_2}. \tag{6}$$

The points of contact on the circle are given by

$$\mathbf{q_i} = \mathbf{u} + r\,\frac{\mathbf{n_i}}{\|\mathbf{n_i}\|}, \qquad i = 1, 2. \tag{7}$$

Tangency condition:

For tangents from an external point **P** to the circle, we have

$$\mathbf{v}^\top \frac{\mathbf{n}}{||\mathbf{n}||} = r. \tag{8}$$

Substituting (6) into (8), and noting that $\mathbf{v} = \sqrt{s}\,\mathbf{u_1}$ and $\mathbf{v}^\top \mathbf{u_2} = 0$, we get

$$\alpha\sqrt{s} = r \quad \Rightarrow \quad \boxed{\alpha = \frac{r}{\sqrt{s}}.} \tag{9}$$

Unit length condition:

$$\alpha^2 + \beta^2 = 1. \tag{10}$$

$$\beta^2 = 1 - \frac{r^2}{s} \quad \Rightarrow \quad \boxed{\beta = \pm\sqrt{1 - \frac{r^2}{s}}.} \tag{11}$$

Hence,

$$\boxed{\frac{\mathbf{n}}{||\mathbf{n}||} = \frac{r}{\sqrt{s}}\mathbf{u_1} \pm \sqrt{1 - \frac{r^2}{s}}\,\mathbf{u_2}.} \tag{12}$$

$$\Rightarrow \boxed{\mathbf{q_i} = \mathbf{u} + \frac{r^2}{\sqrt{s}}\mathbf{u_1} \pm r\sqrt{1 - \frac{r^2}{s}}\,\mathbf{u_2}.} \tag{13}$$

Given $r = 4$, $\sqrt{s} = 7, \mathbf{u} = \mathbf{0}$,

$$\frac{r^2}{\sqrt{s}} = \frac{16}{7} \tag{14}$$

$$r\sqrt{1 - \frac{r^2}{s}} = 4\sqrt{1 - \frac{16}{49}} = \frac{4\sqrt{33}}{7}. \tag{15}$$

$$\mathbf{q_1} = \begin{pmatrix} \frac{16}{7} \\ \frac{4\sqrt{33}}{7} \end{pmatrix}, \quad \mathbf{q_2} = \begin{pmatrix} \frac{16}{7} \\ -\frac{4\sqrt{33}}{7} \end{pmatrix}. \tag{16}$$

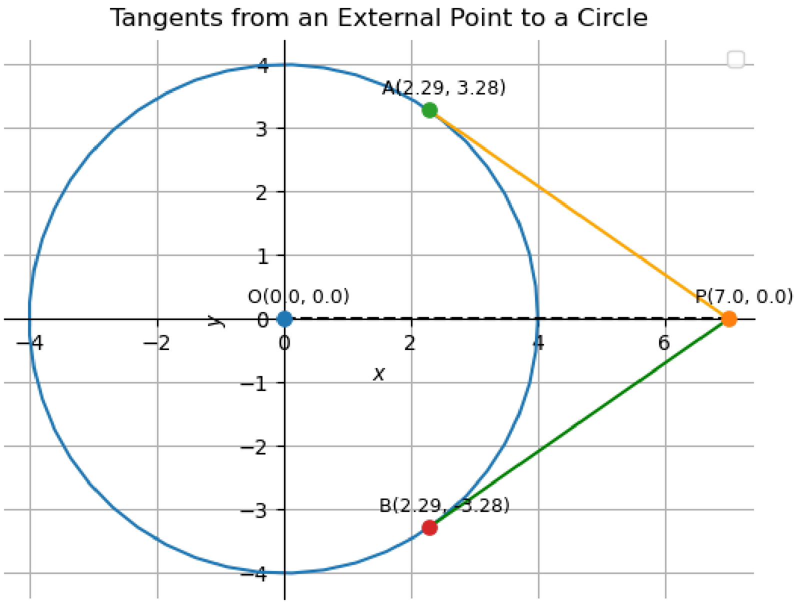
Tangents from an External Point to a Circle
A(2.29, 3.28)
O(0.0, 0.0)
P(7.0, 0.0)
B(2.29, -3.28)
x
y
−4
−2
0
2
4
6
4
3
2
1
0
−1
−2
−3
−4

Figure 1